

Package ‘censoredCI’

June 30, 2026

Type Package

Title Maximum Likelihood Estimation for Type-II and Progressive Censored Data

Version 0.1.0

Description Provides maximum likelihood estimation and asymptotic confidence intervals for user-defined lifetime distributions under Type-II, progressive Type-II, and progressive first-failure censoring schemes. The package implements three main functions, ``ttcs()`` for left, right, and double Type-II censored data, ``pttcs()`` for progressive Type-II censored data, and ``pffcs()`` for progressive first-failure censored data. Users provide observed failure times, initial parameter values, relevant censoring information, and user-defined probability density and cumulative distribution functions. For progressive censoring models, the censoring scheme is supplied directly by the user through a numeric vector. The package supports classical local optimization methods such as 'BFGS' and 'Brent', together with global and population-based metaheuristic algorithms including 'Genetic Algorithm (GA)', 'Differential Evolution (DE)', 'Particle Swarm Optimization (PSO)', 'Grey Wolf Optimizer (GWO)', 'Whale Optimization Algorithm (WOA)', 'Moth-Flame Optimization (MFO)', 'Sine Cosine Algorithm (SCA)', 'Bat Algorithm (BA)', 'Black Hole Optimization (BHO)', 'Cuckoo Optimization Algorithm (COA)' and 'Clonal Selection Algorithm (CLG)'. When automatic method selection is requested, candidate algorithms are compared according to their attained negative log-likelihood values and the best-performing method is reported. The package returns parameter estimates, variance estimates, standard errors, asymptotic confidence intervals, log-likelihood value and the selected optimization method.

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URL <https://github.com/receptanii/censoredCI>

BugReports <https://github.com/receptanii/censoredCI/issues>

Depends R (>= 3.5)

Imports DEoptim,
GA,
MASS,
metaheuristicOpt,
pso,
stats

```
Suggests knitr,
          rmarkdown,
          testthat (>= 3.0.0)
VignetteBuilder knitr
Config/roxygen2/version 8.0.0
Config/testthat/edition 3
Encoding UTF-8
LazyData true
Roxygen list(markdown = TRUE)
RoxygenNote 8.0.0
```

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pffcs	<i>Inference for Progressive First Failure Censoring Schemes</i>
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Description

pffcs computes the Maximum Likelihood Estimates (MLE), variances, standard errors, and asymptotic 95\ of any user-defined lifetime distribution under a Progressive First-Failure Censoring scheme.

Usage

```
pffcs(x_obs, R, par_start, pdf_f, cdf_f, opt_method = "auto", maxiter = 200)
```

Arguments

x_obs	A numeric vector containing the completely observed failure times (ordered observations).
R	A non-negative integer vector specifying the progressive censoring scheme (number of units removed at each failure time). Must have the same length as x_obs.
par_start	A numeric vector of initial parameter values for the optimization process.
pdf_f	A user-defined function calculating the Probability Density Function (PDF) of the distribution. Must accept arguments (x, par).
cdf_f	A user-defined function calculating the Cumulative Distribution Function (CDF) of the distribution. Must accept arguments (x, par).
opt_method	A character string specifying the optimization method. Options include "auto" (runs all available algorithms), "BFGS", "GA", "DE", "PSO", etc. Default is "auto".
maxiter	An integer indicating the maximum number of iterations or generations for the meta-heuristic optimizers. Default is 200.

Details

The function transforms parameters into the logarithmic scale during optimization to ensure positivity constraints. Standard errors are computed using the inverse of the Hessian matrix, with a fallback to the Moore-Penrose pseudoinverse if inversion fails.

Value

A list containing three distinct components:

- **estimates**: A data frame displaying the estimated parameters, their variances, standard errors, 95\
- **stats**: A named numeric vector containing the maximized Log-Likelihood value (LogLik), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).
- **optimizer**: A character string stating the name of the best-performing optimization algorithm.

References

- Aarset, M. V. (1987). How to identify a bathtub hazard rate. *IEEE Transactions on Reliability*, 36(1), 106-108.
- Murthy, D. P., Xie, M., & Jiang, R. (2004). *Weibull models*. John Wiley & Sons.
- Ijaz, M., Mashwani, W. K., & Belhaouari, S. B. (2020). An innovative family in the lifetime distribution field with applications to real and simulated data. *PLOS ONE*, 15(10), e0238746.

Examples

```
# Example using a standard two-parameter Weibull distribution
my_data <- c(0.040, 1.866, 2.385, 3.443, 0.301, 1.876, 2.481, 3.467, 0.309, 1.899,
            2.610, 3.478, 0.557, 1.911, 2.625, 3.578, 0.943, 1.912, 2.632, 3.595,
            1.070, 1.914, 2.646, 3.699, 1.124, 1.981, 2.661, 3.779, 1.248, 2.010,
            2.688, 3.924, 1.281, 2.038, 2.823, 4.035, 1.281, 2.085, 2.890, 4.121,
            1.303, 2.089, 2.902, 4.167, 1.432, 2.097, 2.934, 4.240, 1.480, 2.135,
            2.962, 4.255, 1.505, 2.154, 2.964, 4.278, 1.506, 2.190, 3.000, 4.305,
            1.568, 2.194, 3.103, 4.376, 1.615, 2.223, 3.114, 4.449, 1.619, 2.224,
            3.117, 4.485, 1.652, 2.229, 3.166, 4.570, 1.652, 2.300, 3.344, 4.602,
            1.757, 2.324, 3.376, 4.663)

# Define PDF and CDF in the User Section
user_pdf <- function(x, par) {
  (par[2]/par[1]) * (x/par[1])^(par[2]-1) * exp(-(x/par[1])^par[2])
}
user_cdf <- function(x, par) {
  1 - exp(-(x/par[1])^par[2])
}

user_scheme <- c(2, 0, 1, 0, 0, 3, 0, 0, 1, 0, 1, 0)
sorted_data <- sort(my_data)
obs_data <- sorted_data[1:12]
user_par_start <- c(2,3)

# Run the Progressive First Failure Censoring Inference
result <- pffcs(x_obs = obs_data,
               R = user_scheme,
               par_start = user_par_start,
               pdf_f = user_pdf,
```

```

      cdf_f      = user_cdf,
      opt_method = "auto")

print(result$estimates)

```

pttcs

Inference for Progressive Type-II Censoring Schemes

Description

pttcs computes the Maximum Likelihood Estimates (MLE), variances, standard errors, and asymptotic 95% of any user-defined lifetime distribution under a Progressive Type-II Censoring scheme.

Usage

```
pttcs(x_obs, R, par_start, pdf_f, cdf_f, opt_method = "auto", maxiter = 200)
```

Arguments

x_obs	A numeric vector containing the completely observed failure times (ordered observations).
R	A non-negative integer vector specifying the progressive censoring scheme (number of units removed at each failure time). Must have the same length as x_obs.
par_start	A numeric vector of initial parameter values for the optimization process.
pdf_f	A user-defined function calculating the Probability Density Function (PDF) of the distribution. Must accept arguments (x, par).
cdf_f	A user-defined function calculating the Cumulative Distribution Function (CDF) of the distribution. Must accept arguments (x, par).
opt_method	A character string specifying the optimization method. Options include "auto" (runs all available algorithms), "BFGS", "GA", "DE", "PSO", etc. Default is "auto".
maxiter	An integer indicating the maximum number of iterations or generations for the meta-heuristic optimizers. Default is 200.

Details

The function transforms parameters into the logarithmic scale during optimization to ensure positivity constraints. Standard errors are computed using the inverse of the Hessian matrix, with a fallback to the Moore-Penrose pseudoinverse if inversion fails.

Value

A list containing three distinct components:

- **estimates:** A data frame displaying the estimated parameters, their variances, standard errors, 95%
- **stats:** A named numeric vector containing the maximized Log-Likelihood value (LogLik), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).
- **optimizer:** A character string stating the name of the best-performing optimization algorithm.

References

- Aarset, M. V. (1987). How to identify a bathtub hazard rate. *IEEE Transactions on Reliability*, 36(1), 106-108.
- Murthy, D. P., Xie, M., & Jiang, R. (2004). *Weibull models*. John Wiley & Sons.
- Ijaz, M., Mashwani, W. K., & Belhaouari, S. B. (2020). An innovative family in the lifetime distribution field with applications to real and simulated data. *PLOS ONE*, 15(10), e0238746.

Examples

```
# Example using a standard two-parameter Weibull distribution
my_data <- c(0.040, 1.866, 2.385, 3.443, 0.301, 1.876, 2.481, 3.467, 0.309, 1.899,
            2.610, 3.478, 0.557, 1.911, 2.625, 3.578, 0.943, 1.912, 2.632, 3.595,
            1.070, 1.914, 2.646, 3.699, 1.124, 1.981, 2.661, 3.779, 1.248, 2.010,
            2.688, 3.924, 1.281, 2.038, 2.823, 4.035, 1.281, 2.085, 2.890, 4.121,
            1.303, 2.089, 2.902, 4.167, 1.432, 2.097, 2.934, 4.240, 1.480, 2.135,
            2.962, 4.255, 1.505, 2.154, 2.964, 4.278, 1.506, 2.190, 3.000, 4.305,
            1.568, 2.194, 3.103, 4.376, 1.615, 2.223, 3.114, 4.449, 1.619, 2.224,
            3.117, 4.485, 1.652, 2.229, 3.166, 4.570, 1.652, 2.300, 3.344, 4.602,
            1.757, 2.324, 3.376, 4.663)

# Define PDF and CDF in the User Section
user_pdf <- function(x, par) {
  (par[2]/par[1]) * (x/par[1])^(par[2]-1) * exp(-(x/par[1])^par[2])
}
user_cdf <- function(x, par) {
  1 - exp(-(x/par[1])^par[2])
}

user_scheme <- c(2, 0, 1, 0, 0, 3, 0, 0, 1, 0, 1, 0)
sorted_data <- sort(my_data)
obs_data <- sorted_data[1:12]
user_par_start <- c(2,3)

# Run the Progressive Type-II Censoring Inference
result <- ptcs(x_obs = obs_data,
               R      = user_scheme,
               par_start = user_par_start,
               pdf_f    = user_pdf,
               cdf_f    = user_cdf,
               opt_method = "auto")

print(result$estimates)
```

Description

ttcs Computes the Maximum Likelihood Estimates (MLE), variances, standard errors, and asymptotic 95% of any user-defined lifetime distribution under a Type-II Censoring scheme.

Usage

```
ttcs(
  x,
  par_start,
  pdf_f,
  cdf_f,
  r1,
  r2,
  censoring = "double",
  optimizer = "auto"
)
```

Arguments

<code>x</code>	A numeric vector containing the observed data points.
<code>par_start</code>	A numeric vector indicating the initial starting values for the parameters.
<code>pdf_f</code>	A user-defined R function that represents the analytical Probability Density Function (PDF).
<code>cdf_f</code>	A user-defined R function that represents the analytical Cumulative Distribution Function (CDF).
<code>r1</code>	An integer specifying the number of left-censored observations (set to 0 if not left-censored).
<code>r2</code>	An integer specifying the number of right-censored observations (set to 0 if not right-censored).
<code>censoring</code>	A character string defining the censoring architecture. Must be one of "left", "right", or "double". Default is "double".
<code>optimizer</code>	A character string selecting the optimization strategy. If set to "auto", all available global metaheuristics are evaluated. Default is "auto".

Details

The parameters are estimated by maximizing the censored log-likelihood function. To guarantee that the scale and shape parameters remain strictly positive during numerical optimization, the internal objective function optimizes the parameters on the log-scale and maps them back using the exponential function. Standard errors are derived from the numerical Hessian matrix evaluated at the joint MLEs.

Value

A list containing three distinct components:

- **estimates:** A data frame displaying the estimated parameters, their variances, standard errors, 95\
- **stats:** A named numeric vector containing the maximized Log-Likelihood value (LogLik), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).
- **optimizer:** A character string stating the name of the best-performing optimization algorithm.

References

- Aarset, M. V. (1987). How to identify a bathtub hazard rate. *IEEE Transactions on Reliability*, 36(1), 106-108.
- Murthy, D. P., Xie, M., & Jiang, R. (2004). *Weibull models*. John Wiley & Sons.
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```
# Example using a standard two-parameter Weibull distribution
my_data <- c(0.040, 1.866, 2.385, 3.443, 0.301, 1.876, 2.481, 3.467, 0.309, 1.899,
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            1.070, 1.914, 2.646, 3.699, 1.124, 1.981, 2.661, 3.779, 1.248, 2.010,
            2.688, 3.924, 1.281, 2.038, 2.823, 4.035, 1.281, 2.085, 2.890, 4.121,
            1.303, 2.089, 2.902, 4.167, 1.432, 2.097, 2.934, 4.240, 1.480, 2.135,
            2.962, 4.255, 1.505, 2.154, 2.964, 4.278, 1.506, 2.190, 3.000, 4.305,
            1.568, 2.194, 3.103, 4.376, 1.615, 2.223, 3.114, 4.449, 1.619, 2.224,
            3.117, 4.485, 1.652, 2.229, 3.166, 4.570, 1.652, 2.300, 3.344, 4.602,
            1.757, 2.324, 3.376, 4.663)

# Define PDF and CDF in the User Section
user_pdf <- function(x, par) {
  (par[2]/par[1]) * (x/par[1])^(par[2]-1) * exp(-(x/par[1])^par[2])
}
user_cdf <- function(x, par) {
  1 - exp(-(x/par[1])^par[2])
}

# Run the Type-II Censoring Inference
result <- ttcs(x = my_data,
              par_start = c(.2,.2),
              pdf_f = user_pdf,
              cdf_f = user_cdf,
              r1 = 1,
              r2 = 1,
              censoring = "double",
              optimizer = "auto")

print(result$estimates)
```

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